

# JOONWON LEE

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## SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic modeling, scalable likelihood-based inference, simulation, statistical diagnostics, and end-to-end model validation. Builds reproducible Python/PyTorch workflows for large, time-dependent datasets, integrates compiled C++ components for performance-critical routines, and translates analytical problems into testable model specifications.

## EDUCATION

**Ph.D. in Statistics, Rutgers University, Piscataway, NJ**

Expected Jan 2027 | GPA: 3.8/4.0

**M.S. in Statistics, University of Minnesota, Minneapolis, MN**

Sep 2021 | GPA: 3.7/4.0

## TECHNICAL SKILLS

**Programming and Platforms:** Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable analytical pipelines

**Quantitative Methods:** Stochastic processes, time-series and spatiotemporal modeling, Gaussian processes, MLE/GLS, Monte Carlo simulation, numerical optimization, regression and machine learning, causal inference, hypothesis testing, spectral analysis, model selection, diagnostics, and validation

## EXPERIENCE

**JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review**

Summer 2026

- Developed and validated a Sequential Probability Ratio Test framework for binary risk indicators, translating a monitoring requirement into explicit hypotheses, likelihood-ratio boundaries, and statistically controlled decision rules.
- Applied Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping-time behavior, Type I/II error, and model limitations.

**Travelers - Data Science Leadership Program, Business Insurance Pricing Team**

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

## QUANTITATIVE RESEARCH AND PROJECTS

**Ph.D. Dissertation Research - Scalable Stochastic Modeling and Diagnostics**

Sep 2024 - Present

- **Scalable Time-Dependent Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, using ordered conditional models to fit up to 145,008 observations per day.
- **Simulation and Model Diagnostics:** Designed controlled simulation studies and scale- and frequency-resolved diagnostics to identify misspecification in low-frequency structure, high-frequency behavior, noise, and dependence.
- **Data and Model Lifecycle:** Built workflows for quality filtering, missingness tracking, time-dependent grid alignment, regular-grid construction, estimation, optimization, sensitivity analysis, cached outputs, and restartable HPC execution.
- **Research Engineering:** Integrated compiled C++ max-min ordering routines into Python through pybind11 and implemented reusable Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, and CPU/GPU execution.

**Econometrics and Causal Inference Project - Earned Income Tax Credit and Labor Supply**

Graduate Research

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching, with explicit treatment, outcome, comparison-group, and confounding assumptions.

## ADDITIONAL

**Coursework:** Econometrics, Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

**Languages:** Korean (Native); English (Fluent); Chinese (Basic)